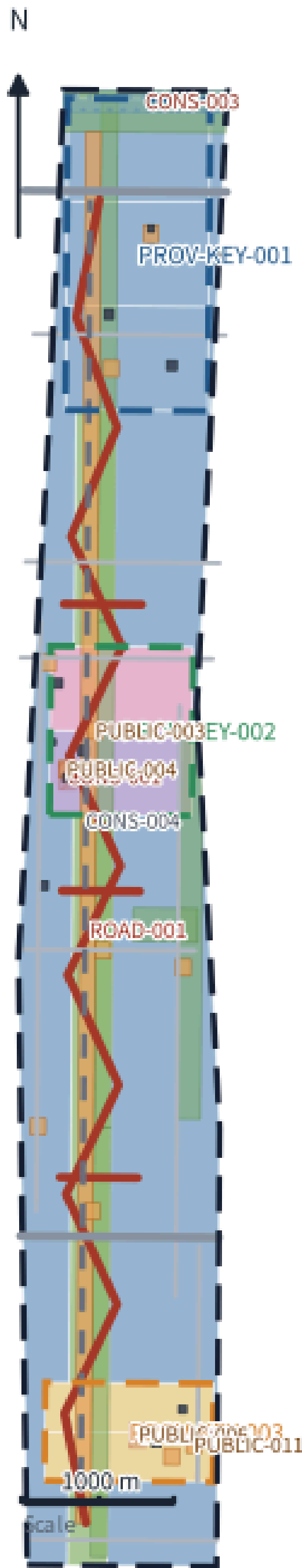


OVERALL SPATIAL STRUCTURE & LOCATION PLAN • THE JINGZHANG DIAGRAM (provisional)

Base: this package's geometry/\*.geojson nine layers • EPSG:4548 • north arrow, scale bar, legend, feature\_id labels

PROVISIONAL BOUNDARY (NON-STATUTORY)



Legend (layers & feature\_id)

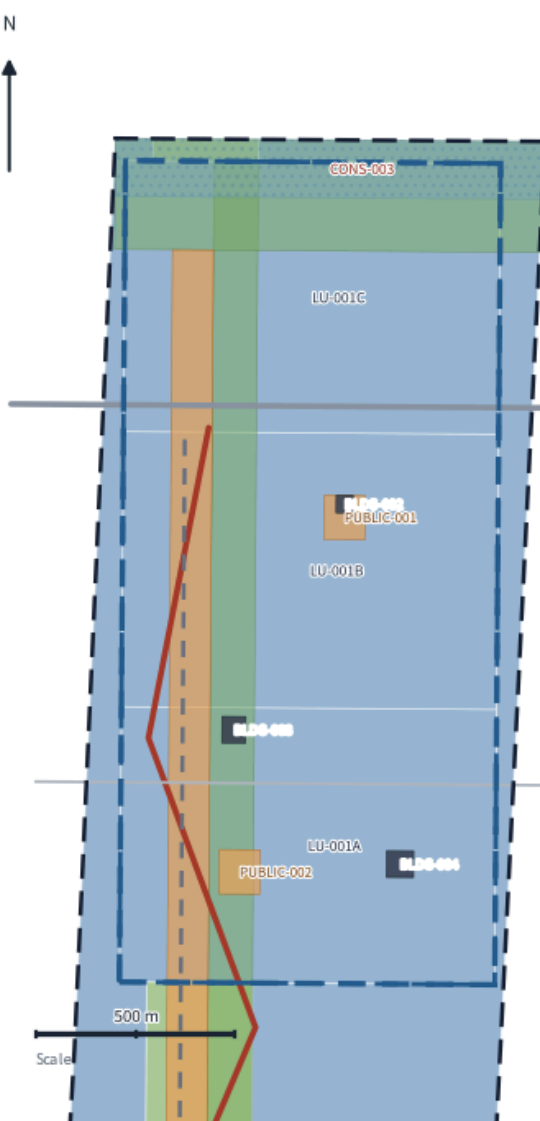
- Site boundary SITE-001 (provisional)
- Key areas PROV-KEY-001/002/003
- Land use LU-001A~LU-005 (colored by code)
- Green space GREEN-001~006
- Public space PUBLIC-000~012
- Building footprints BLDG-001~010
- Road centerlines ROAD-001~014 (greenway/arterial/expressway)
- Constraints CONS-001~004 (heritage/blue line/rail, indicative)

Drawn from the organizer's provisional geometry; not an official redline. Areas and alignments follow metrics.json recomputation and the GeoJSON layers; redraw in full once official data is released.

ZHONGZHIYUAN AI ACCELERATION AREA PLAN (Test Track • provisional)

Land-use bands • building footprints • public nodes • heritage setbacks • feature\_id labels (viewport = key area + 350 m)

PROVISIONAL BOUNDARY (NON-STATUTORY)



- Legend (layers & feature\_id)
- Land use LU-001A~LU-005 (colored by code)
  - Green space GREEN-001~006
  - Public space PUBLIC-000~012
  - Building footprints BLDG-001~010
  - Road centerlines ROAD-001~014 (greenway/arterial/expressway)
  - Constraints CONS-001~004 (heritage/blue line/rail, indicative)
  - Site boundary SITE-001 (provisional)

Drawn from the organizer's provisional geometry; not an official redline. Areas and alignments follow metrics.json recomputation and the GeoJSON layers; redraw in full once official data is released.

BEIJING AI ORIGIN COMMUNITY PLAN (Departure Yard • provisional)

Land-use bands • building footprints • public nodes • heritage setbacks • feature\_id labels (viewport = key area + 350 m)

PROVISIONAL BOUNDARY (NON-STATUTORY)



- Legend (layers & feature\_id)
- Land use LU-001A~LU-005 (colored by code)
  - Green space GREEN-001~006
  - Public space PUBLIC-000~012
  - Building footprints BLDG-001~010
  - Road centerlines ROAD-001~014 (greenway/arterial/expressway)
  - Constraints CONS-001~004 (heritage/blue line/rail, indicative)
  - Site boundary SITE-001 (provisional)

Drawn from the organizer's provisional geometry; not an official redline. Areas and alignments follow metrics.json recomputation and the GeoJSON layers; redraw in full once official data is released.

DAZHONGSI AI INDUSTRY CLUSTER PLAN (Marshalling Yard • provisional)

Land-use bands • building footprints • public nodes • heritage setbacks • feature\_id labels (viewport = key area + 350 m)

PROVISIONAL BOUNDARY (NON-STATUTORY)



- Legend (layers & feature\_id)
- Land use LU-001A~LU-005 (colored by code)
  - Green space GREEN-001~006
  - Public space PUBLIC-000~012
  - Building footprints BLDG-001~010
  - Road centerlines ROAD-001~014 (greenway/arterial/expressway)
  - Constraints CONS-001~004 (heritage/blue line/rail, indicative)
  - Site boundary SITE-001 (provisional)

Drawn from the organizer's provisional geometry; not an official redline. Areas and alignments follow metrics.json recomputation and the GeoJSON layers; redraw in full once official data is released.



## THE JINGZHANG DIAGRAM: OVERALL SPATIAL OPERATING SYSTEM

Four Epochs of Innovation along the Corridor & Three Key Areas of the Spatial OS

PROVISIONAL BOUNDARY (NON-STATUTORY)

### I. FOUR INNOVATION EPOCHS

#### 1909 Jingzhang Railway

First trunk railway independently surveyed, designed and built by China; switchback topology beats the grade  
[National Railway Museum Records]

#### 1980s Zhongguancun

Historical narrative milestone of Chinese tech entrepreneurship and institutional innovation  
[Historical Narrative Milestone]

#### 2019 Jingzhang High-Speed Railway

World's first 350 km/h intelligent high-speed railway (full qualifier); intelligent running & Beidou at 350 km/h  
[Ministry of Transport Official Statement]

#### 2026 Centennial AI Innovation Belt

Proposal design proposition: human-AI co-ascent; diagram governance & The Human Switch safety  
[Design Concept Proposition]

### II. SPATIAL OPERATING SYSTEM - THREE KEY AREAS

#### Zhongzhiyuan AI Testbed

192.1 ha • Microgrid compute  
Autonomous loop test track

#### Beijing AI Origin Community

104.3 ha • Staffed dispatch hall  
Living district & night school

#### Dazhongsi Marshalling Yard

72.0 ha • Commercialisation  
Two-part handover desk

Provisional boundary polygon • EPSG:4548 model internal only

Provisional boundary polygon • EPSG:4548 model internal only

Provisional boundary polygon • EPSG:4548 model internal only

#### Four industry test & validation scenarios:

Autonomous safety corridor

Robot city-service yard

LLM governance sandbox

Heritage digital twin

### III. DIAGRAM GOVERNANCE: 4 HARDWIRED INTERLOCK SYSTEMS

#### 信号制 Signal

Multi-tiered dynamic permissions  
Charter 10 human-first principles

#### 闭塞制 Block

Spatial & compute separation  
Charter 10 human-first principles

#### 联锁制 Interlock

SEB v0.5.0 tabletop self-check 7/7 (participant-declared)  
Charter 10 human-first principles

#### 路票制 Token

Human Switch tiered interface & opt-out  
Charter 10 human-first principles

submissions/Juwan-Hwang/jingzhang-diagram • geometry/\*.geojson • metrics.json • 图表为过程性研究表达，非法定依据

## SLOW MOBILITY, TRANSIT & BLUE-GREEN PUBLIC SPACE NETWORK

≈9.5 km spine + stitch corridors ≈ 11.0 km slow network (recomputed) • Pedestrian Priority • The Human Switch (Tiered Safety Interface)

PROVISIONAL GEOMETRY • EPSG:4548

### 1. MULTI-TIERED MOBILITY NETWORK

#### Spine Continuous Greenway

≈9.5 km linear spine (recomputed) linking Beijing North to Tsinghua

#### AI Micro-Shuttle Test Lane

Speed cap < 15 km/h, dedicated lane, pedestrian priority

#### Barrier-Free & Silver Walkways

Full tactile paving & gentle slopes (design-floor target), emergency call points

#### Ring-Road Stitching Bridges

Bridging Ring 2/3/4 barriers for uninterrupted mobility

### 2. BLUE-GREEN SYSTEM & THE HUMAN SWITCH

#### Xiaoyuehe Sponge Green Infrastructure

178.94 ha green space, stormwater retention, microclimate

#### Origin Civic Plazas & Open Space

106.23 ha public space for community life and events

#### The Human Switch — Tiered Safety Interface

Public alert logged & displayed -> authorised on-site e-stop (long press + two-person confirm) -> professional recovery at dispatch hall; fully audited with accountability chain (conceptual; parameters are design assumptions pending field testing)

#### SEB v0.5.0 Hardwired Self-Check

Two-part handover ledger, staffed windows; 7/7 tabletop fixtures match (participant-declared, synthetic fixtures, no third-party certification)

## QUANTITATIVE METRICS, SIMULATION LEDGER & EVIDENCE CROSSWALK

Spatial Recalculation • 10 Synthetic Tasks (90% Success) • SEB v0.5.0 Tabletop Self-Check (7/7 Match) • 7-Dimension Rubric

PROVISIONAL GEOMETRY • EPSG:4548

### 1. SPATIAL & SIMULATION METRICS

| Metric Key                                | Declared Value                | Provenance / Formula                                                     |
|-------------------------------------------|-------------------------------|--------------------------------------------------------------------------|
| Total Site Area (site_area_sqm)           | 11,412,825.39 m <sup>2</sup>  | EPSG:4548 provisional boundary                                           |
| Green Space Area (green_space_area_sqm)   | 1,789,421.32 m <sup>2</sup>   | unary_union intersection                                                 |
| Green Ratio (green_ratio)                 | 0.15679 (15.68%)              | Exact ratio (topology-consistent)                                        |
| Public Space Area (public_space_area_sqm) | 1,062,301.12 m <sup>2</sup>   | unary_union intersection                                                 |
| Public Space Ratio (public_space_ratio)   | 0.09308 (9.31%)               | Exact ratio (topology-consistent)                                        |
| Simulation Task Count                     | 10 tasks (SIM-001-010)        | Deterministic seed: 20260831                                             |
| Simulation Success Rate                   | 0.90 (90.0%)                  | Honest disclosure of overrun                                             |
| SEB Tabletop Self-Check                   | 7 / 7 fixtures match (Exit 0) | Participant-declared • synthetic fixtures • no third-party certification |

指标与 simulation.json / metrics.json 机器可读值一致；SEB 推演为参与看桌面自检，可离线复跑，未经第三方认证

### 2. RUBRIC CROSSWALK & DATA GAP DISCLOSURE

|                                                                                                                      |
|----------------------------------------------------------------------------------------------------------------------|
| <b>Brief Alignment (20%)</b><br>3-level scopes • 6 required tasks • 3 key areas • ≈11.0 km slow network (recomputed) |
| <b>Originality (10%)</b><br>Working Diagram concept • Switchback unit • 4 innovation epochs                          |
| <b>AI Innovation (15%)</b><br>4-system interlock • 12 scenario cards • SEB v0.5.0 self-check 7/7                     |
| <b>Feasibility (20%)</b><br>JZ-01-10 lifecycle ledger • Adopter rights • 4-phase rollout                             |
| <b>Public Interest (10%)</b><br>Staffed-window floor (design target) • Barrier-free • The Human Switch (tiered)      |
| <b>Risk Compliance (10%)</b><br>Provisional boundaries • Heritage setbacks • Adverse readings                        |
| <b>Completeness (15%)</b><br>Bilingual parity • Embedded fonts • A3/A0 drawings • 4 Gate PASS                        |
| <b>Statutory Notice:</b> Official redlines & FAR are unknown; conceptual design does not replace statutory permits.  |

## LAND USE STRUCTURE & ZONING PARTITION (TOPOLOGY SEAMLESS)

11.41 km<sup>2</sup> Overall Design Scope • Green Ratio 15.68% • Public Space Ratio 9.31% • 9 polygons seamless tiling

PROVISIONAL GEOMETRY • EPSG:4548

### 1. LAND-USE FUNCTIONAL MIX (recomputed from land\_use.geojson)

|                                                                                                                                                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>0802 Research &amp; two-wing tech services</b><br>808.59 ha • ≈70.8%<br>LU-001A/B/C + LU-005 (Zhongzhiyuan test track & wing interfaces)          |
| <b>1401 Park green spine</b><br>156.33 ha • ≈13.7%<br>LU-004 (Jingzhang heritage park blue-green open space)                                         |
| <b>05 Commercial services</b><br>72.05 ha • ≈6.3%<br>LU-003A/B (Dazhongsi AI-native market & station services)                                       |
| <b>0803 Culture</b><br>52.16 ha • ≈4.6%<br>LU-002A (Origin cultural core: hall / plaza / dispatch)                                                   |
| <b>0702 Community services</b><br>52.16 ha • ≈4.6%<br>LU-002B (incubation & scholar-alleys mixed)                                                    |
| <b>Slow-mobility network (centerlines)</b><br>≈9.5 km spine • ≈11.0 km total<br>ROAD-001 + stitch corridors ROAD-012/013/014 (not land-use polygons) |

面积基于 land\_use.geojson 各图斑 area\_sqm\_declared (EPSG:4548)；慢行里程为 roads.geojson greenway 中心线复算值 [metric:slow\_corridor\_length\_m]

### 2. SPATIAL METRIC RECALCULATION

|                                                                                                                                                                                                                     |                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| Total Site Area                                                                                                                                                                                                     | 11,412,825.39 m <sup>2</sup> (11.41 km <sup>2</sup> ) |
| Green Space Union Area                                                                                                                                                                                              | 1,789,421.32 m <sup>2</sup> (178.94 ha)               |
| Green Ratio                                                                                                                                                                                                         | 0.15679 (15.68%)                                      |
| Public Space Union Area                                                                                                                                                                                             | 1,062,301.12 m <sup>2</sup> (106.23 ha)               |
| Public Space Ratio                                                                                                                                                                                                  | 0.09308 (9.31%)                                       |
| Slow corridor length                                                                                                                                                                                                | 11,005.9 m (recomputed)                               |
| Key Areas Total                                                                                                                                                                                                     | 368.40 ha (3 Key Areas)                               |
| Floor Area Ratio (FAR)                                                                                                                                                                                              | unknown (pending statutory plan)                      |
| Topology Integrity                                                                                                                                                                                                  | PASS (0 gaps / 0 overlaps)                            |
| Design judgment: innovation functions dominate; life & public services form the ≈29.2% base; the two-wing service land holds the largest share and its public-service mix must be subdivided at control-plan stage. |                                                       |